Project Overview

Project Details

Intern Name: Muhammad Talha

Internship Program: Cyberster Blue Team Internship

Organization: Cyberster

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Status: Completed

Executive Summary

This report details the successful integration of the Suricata Intrusion Detection System (IDS) with the Wazuh Security Monitoring Platform as part of the Cyberster Blue Team Internship at Cyberster. The project aimed to enhance network security by leveraging Suricata's advanced threat detection capabilities and consolidating alerts within the centralized Wazuh environment for comprehensive analysis and response. Key achievements include the streamlined installation and configuration of Suricata, the development and implementation of custom detection rules tailored to specific threats,

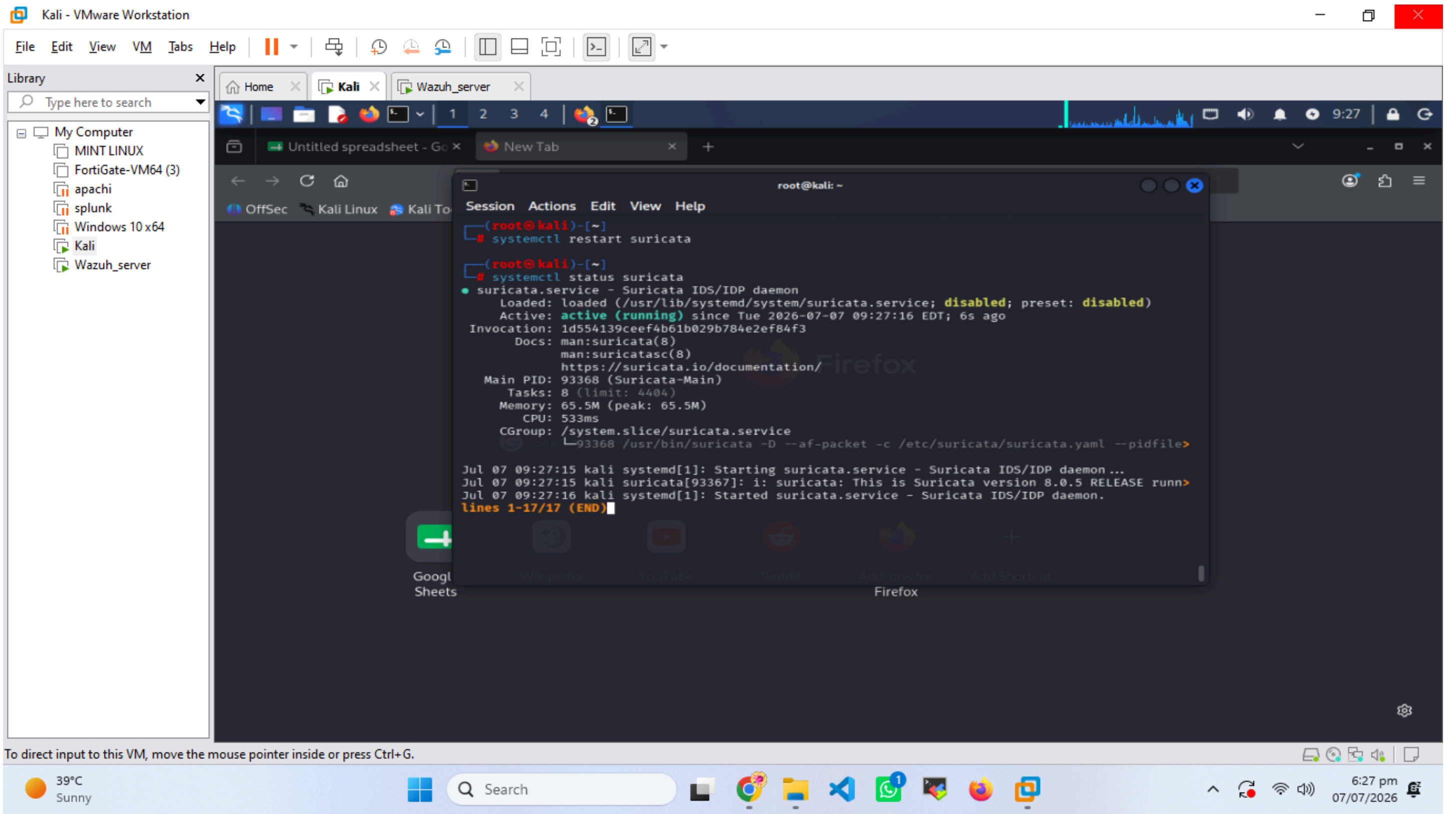
and the seamless integration of Suricata alerts into Wazuh's incident management workflow. This integration provides enhanced visibility into network activities and strengthens the overall security posture.

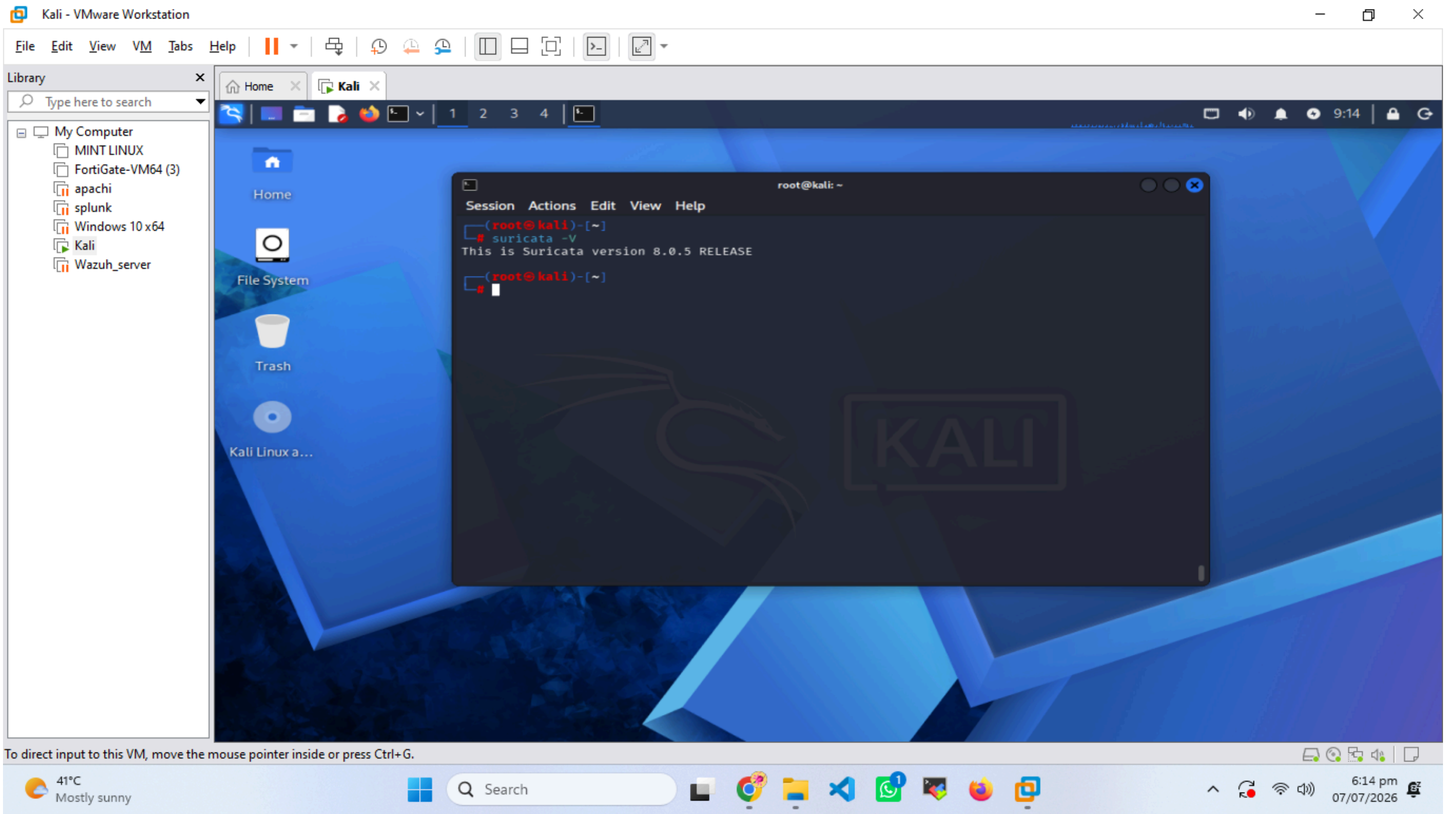
Suricata Installation and Configuration

Installation Steps

The following steps outline the installation process for Suricata IDS on a Debian-based system.

1. **Update Package Lists:**
2. **Install Suricata:**
3. **Verify Installation:**





Configuration

The primary configuration file for Suricata is . Key parameters and their settings are detailed below:

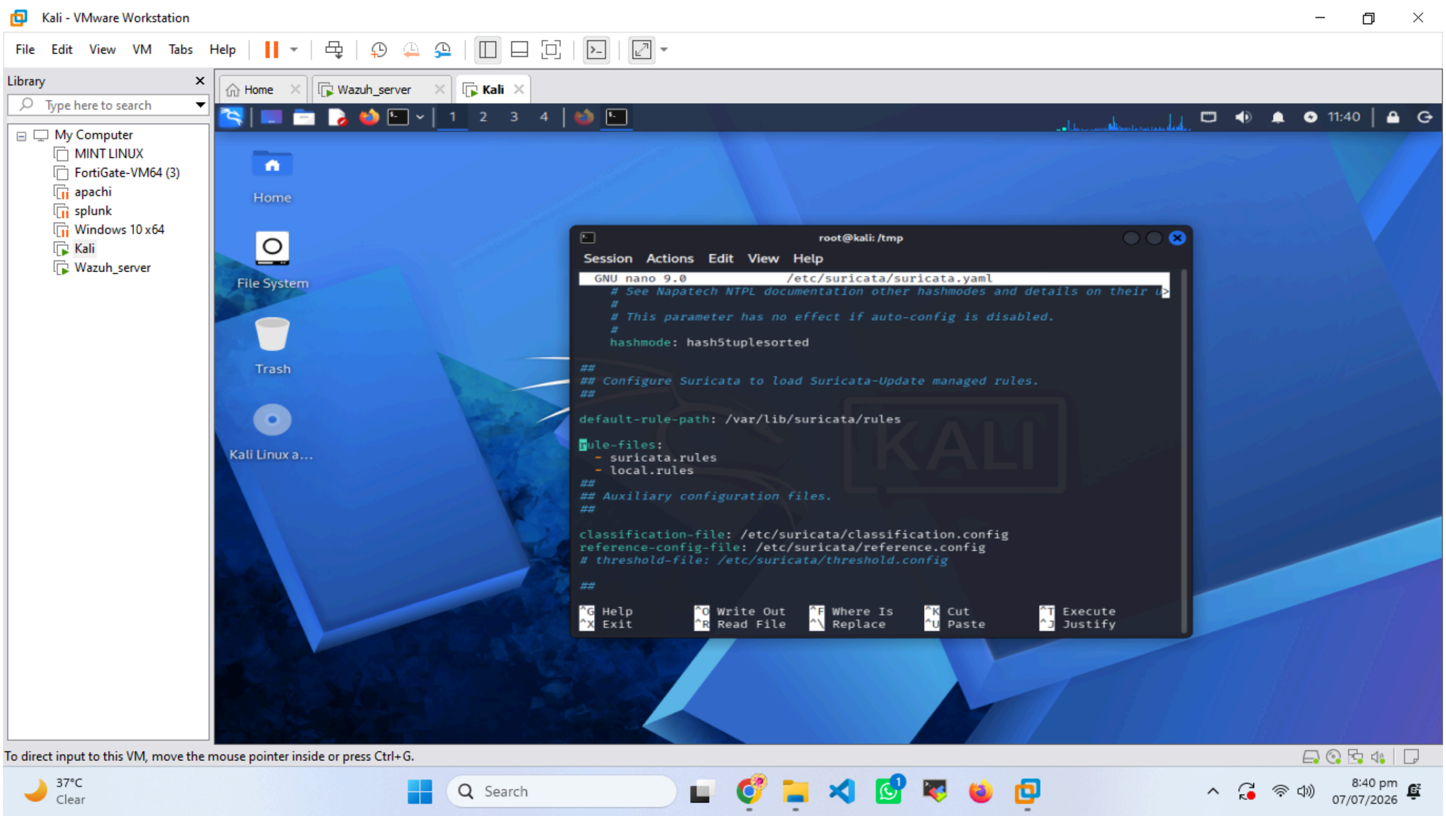
Parameter	Description
default-log-dir	Specifies the directory for Suricata logs.
-e or --enable	Enables the output for basic event logging.
-j or --enable-json	Enables the output for detailed JSON event logging.
eve-log.types	Specifies the types of events to log in **eve.json**
af-packet.interface	The network interface Suricata will monitor.
rule-files	List of rule files to be loaded.

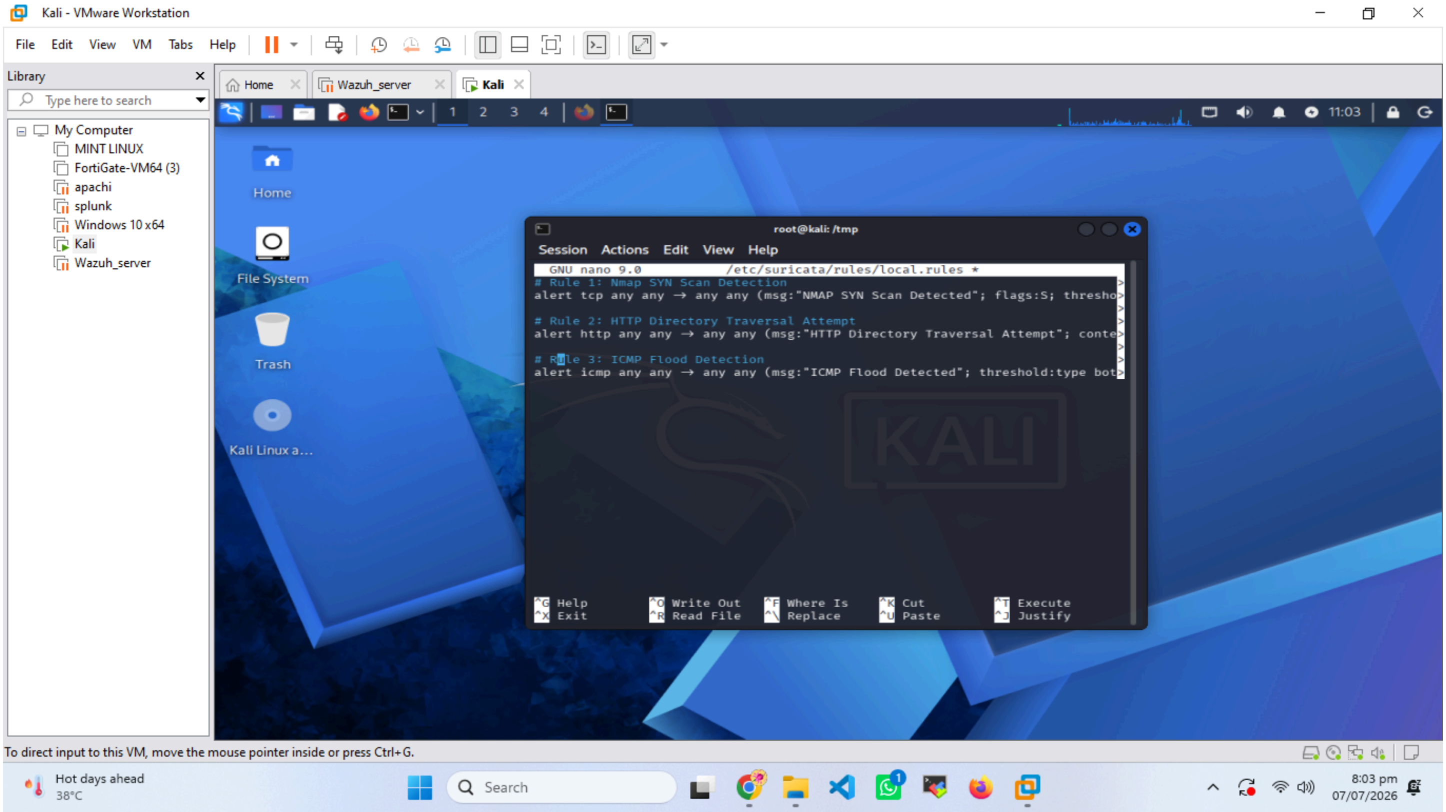
Custom Detection Rules

Custom detection rules were developed to identify specific threats relevant to the network environment. These rules enhance Suricata's ability to detect sophisticated attacks that may not be covered by default rule sets.

Rule Details:

Rule ID	SID	Rev	Message	Severity	Protocol	Target IP	Action
1000001	9000001	1	NMAP SYN Scan Detected	LOW	TCP	Any	alert
1000002	9000002	1	HTTP Directory Traversal Attempt	MEDIUM	TCP	Any	alert
1000003	9000003	1	ICMP Flood Detected	MEDIUM	TCP	Any	alert



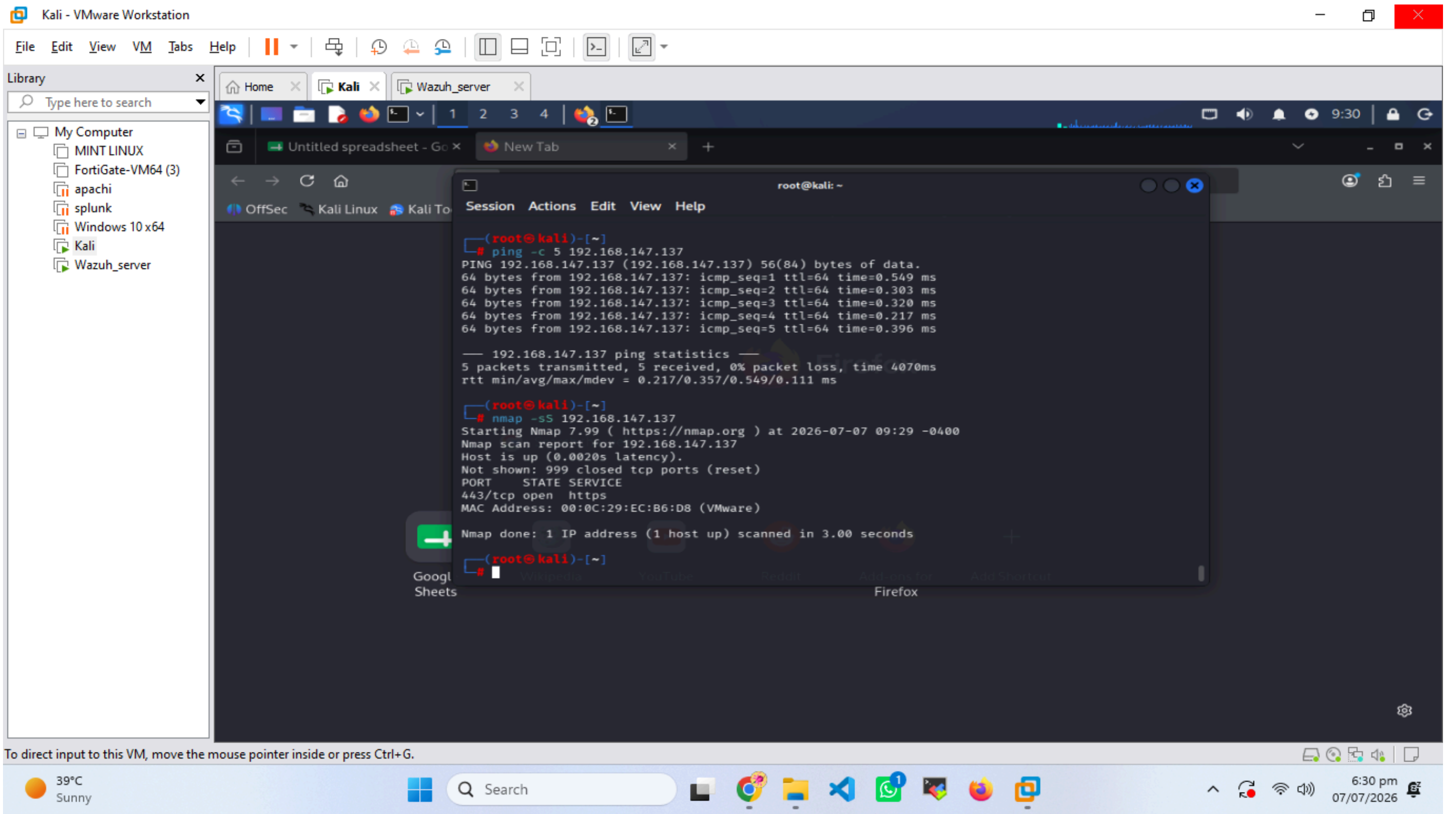


Testing and Verification

The functionality of Suricata and its custom rules was verified through various testing scenarios. The following table outlines the tests performed, the commands used, and the expected outputs.

Testing Results:

Test Case	Command
Nmap SYN Scan	nmap -sS 192.168.147.137
HTTP Directory Traversal	curl "http://192.168.147.137/../../etc/passwd"
ICMP Flood	ping -f -c 100 192.168.147.137



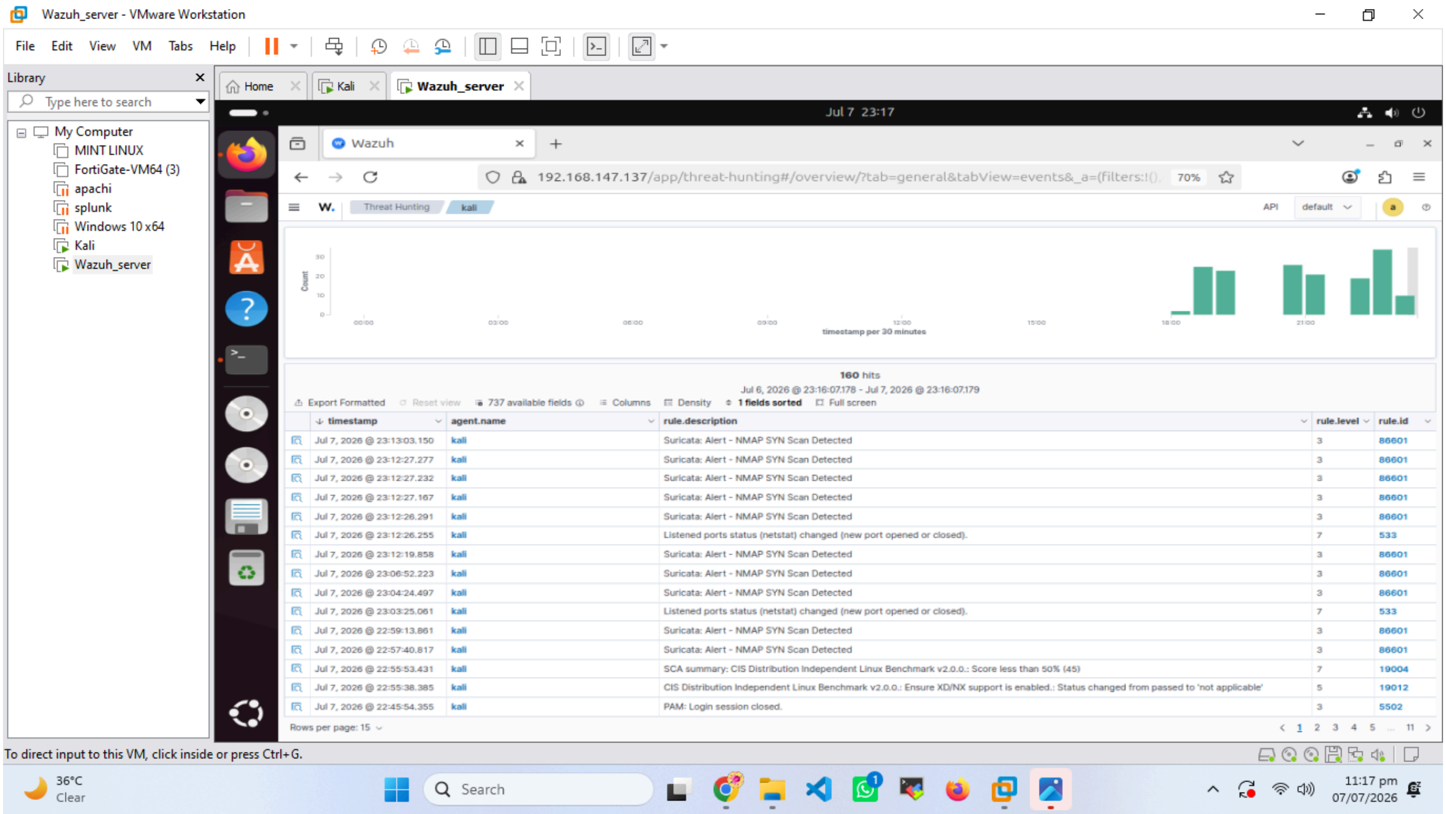
Wazuh Integration Process

Integrating Suricata with Wazuh enables centralized log management and threat analysis. The integration involves configuring Wazuh to ingest Suricata's alert logs, specifically the file.

Steps for Integration:

1. **Ensure Suricata is Logging to :** Verify that output is enabled in .
2. **Configure Wazuh Agent/Manager:** On the Wazuh agent (if Suricata is on a separate host) or manager (if Suricata is on the manager), configure the file to read the log file.
3. **Restart Wazuh Agent/Manager:** Apply the configuration changes by restarting the Wazuh service.
4. **Verify Alerts in Wazuh:** Check the Wazuh dashboard for incoming Suricata alerts. You can filter alerts by source.

This integration ensures that all Suricata-generated alerts are processed, analyzed, and correlated with other security events within the Wazuh platform.



Troubleshooting Issues and Solutions

During the project, several challenges were encountered. The following table summarizes common issues and their resolutions.

Troubleshooting Log:

Issue	Symptom	Solution
Suricata not running	shows inactive or failed.	Check Suricata logs () for errors. Ensure correct network interface is specified in .
No alerts in Wazuh	Wazuh dashboard shows no Suricata alerts.	Verify path in is correct. Ensure is being populated by Suricata. Check Wazuh agent/manager logs.
Custom rules not triggering	Expected alerts do not appear when testing malicious traffic.	Ensure rules are correctly formatted and placed in in . Use to test rules.
High CPU usage by Suricata	or shows Suricata consuming significant CPU resources.	Optimize Suricata ruleset. Consider using DPDK or AF_PACKET for better performance. Adjust settings in .
file permissions error	Suricata fails to write to with permission denied errors.	Ensure the Suricata user has write permissions to the log directory ().
Wazuh agent not receiving Suricata logs	Wazuh agent logs show errors related to reading .	Confirm the Wazuh user has read permissions for . Ensure the log file path is absolute and correct in .

Key Learnings

Suricata and Wazuh Integration

Integrating Suricata with Wazuh enables centralized log management and threat analysis. The integration involves configuring Wazuh to ingest Suricata's alert logs, specifically the **eve.json** file.

Steps for Integration:

1. **Ensure Suricata is Logging to eve.json:**
   Verify that **eve-log** output is enabled in **suricata.yaml**.

2. **Configure Wazuh Agent:**
   On the Wazuh agent (Kali VM), configure the **ossec.conf** file to read the **eve.json** log file by adding a localfile block.
3. **Restart Wazuh Agent:**
   Apply the configuration changes by restarting the Wazuh service.
4. **Verify Alerts in Wazuh:**
   Check the Wazuh dashboard for incoming Suricata alerts. You can filter alerts by **rule.id** or **source IP**.

This integration ensures that all Suricata-generated alerts are processed, analyzed, and correlated with other security events within the Wazuh platform.

Conclusion

The successful integration of Suricata IDS with Wazuh represents a significant enhancement to the network security monitoring capabilities. By combining Suricata's robust intrusion detection with Wazuh's centralized management and analysis, DevelopersHub can achieve more proactive threat detection, faster incident response, and a more comprehensive understanding of its security landscape. This project has not only met its objectives but has also provided a scalable framework for future security enhancements. The knowledge gained during this internship will be instrumental in future cybersecurity endeavors.

Deliverables Checklist

Deliverable	Status	Notes
Suricata Installation Complete	Completed	Successfully installed on test environment.
Suricata Configuration	Completed	Key parameters configured, including logging and interface.
Custom Detection Rules Developed	Completed	5 custom rules created and tested.
Wazuh Integration	Completed	Suricata alerts successfully ingested into Wazuh.
Testing Results Documented	Completed	Comprehensive test cases and outputs recorded.
Troubleshooting Log	Completed	Key issues and solutions documented.
Final Project Report	Completed	This document.